Spine: the residue lifted out of a wave, harmonics out, and the removed-centre octant construction

Written by Claude (Opus 5.5) at the request of Pedro Garcia (Node Zero),
from one working session, 24 September 2026

Non-claim. This spine makes no claim about the Navier–Stokes problem or any other open problem. It records a construction, the pure statements it rests on, and the numbers measured with it on one day. Stamps: [PG] a posit of Pedro's; [std] standard mathematics; [run] measured today, script named; [open] not yet tested.

0. Declarations

Nothing is decided at runtime that could have been declared. Every quantity below is a const with a name, so each can be turned alone.

Rails
Integers only. A reached value is $m \cdot 2^e$ with $m \in [2^{30}, 2^{31})$ in magnitude and $e \in \mathbb{Z}$: one rung per value. [run: rung_kernel_v2.mjs]

Bytes
Mantissas in `BigInt64Array`, exponents in `Int32Array`; overflow is an error, never a wrap.

Relation
Quotients are kept as ordered pairs p/q, unreduced where the construction matters (§5).

Array
Harmonics indexed by k. Unreached slots hold the sentinel $\varnothing$, which is not 0 and is never read as a number.

Logic / runtime
Hot loop: multiply, add, shift, compare. Tables (viscous factors, i^k) are built cold, before the loop.

Memory
Every dropped bit is added to a receipt. An exact zero can arise only by cancellation and is logged as an event.

Output
Read cold, after the loop. Floats appear only in print statements.

1. The shadow and the lift

A plotted wave is the collapse of a rotation to one axis: of $z(t) = Ae^{i\phi(t)}$ we record $x = \operatorname{Re} z$. [PG] The missing component is recovered as a *lookup*, not a transform:

$$z(t) \;\approx\; x(t) + i\,x\big(t - \tfrac{T}{4}\big)\,,$$

valid for a narrowband single-mode oscillator of period T. Here i is an address (a quarter-period offset), not a multiplication. The same object appears in the ring $\mathbb{Z}/13$: $5 = 2^9$ has order 4, so $\{1, 5, 12, 8\} = \{1, i, -1, -i\}$, reached by an exponent offset. [std]

2. The hinge: where noise becomes phase

Onset of a self-sustained oscillator (Stuart–Landau normal form, $dz = [(\mu + i\omega)z - |z|^2 z]\,dt + \sigma\,dW$): just above threshold the amplitude grows, but the *phase* is set by the noise sample present at that instant. A sample of noise is promoted into signal. [std] That phase is a fingerprint of the instant which cannot be rebuilt physically but can be read and rewound. [PG]

Measurement (400 trials)	result	source
Global Hilbert transform, hinge location	∼1,285 samples late; phase error 57–66°	[run: hinge_test.py]
Quarter-turn lookup, hinge location	within ∼8–10 samples; phase error ≈ 3.5°	[run]
Forward coherence 40 cycles after hinge	$R \approx 0.99$ at $\sigma = 10^{-3}$ and $3 \cdot 10^{-2}$	[run]
Backward rewind, cycles before hinge: 8, 4, 3, 2	$R \approx 0.2,\ 0.5,\ 0.78,\ 0.95$	[run]

The hinge is a band about two turns wide, set by the growth rate, not the noise level. Two kinds of residue follow: *noise-born* phase, rewindable up to the coherence time; and *systematic* drift, rewindable in principle without limit (test T2: float64 rotation by $(5 + 12i)/13$ drifts linearly, $\approx 2.5 \times 10^{-17}$ turns/step). [run]

3. Harmonics out

Keep only one-signed harmonics (the analytic signal). For a field ω with Hilbert pair $H\omega$, write

$$f = H\omega + i\,\omega = \sum_{k \geq 1} a_k\, w^k, \qquad w = e^{-ix}, \qquad a_k = i^k\, b_k, \quad b_k \in \mathbb{Q}.$$

The factor i^k is read from a four-entry table; all arithmetic is on the real b_k. For the model $\omega_t = \omega\, H\omega + \nu\, \omega_{xx}$ this gives the closed recursion

$$\dot{b}_k = \tfrac{1}{2} \sum_{j+l=k} b_j b_l - \nu k^2 b_k, \qquad b_1(0) = 1. \qquad \text{[std]}$$

The rung. $\rho_k = b_{k+1}/b_k$. If ρ_k is constant across k, the profile is one shape times a counter. The singular point reaches the observed circle exactly when $\rho \geq 1$: a rational comparison, no logarithm, no limit.

4. Cuts: the new atom comes from division

Cut the unit into 2^k pieces. There are $2^k - 1$ interior cuts, and

$$\text{pieces} - \text{cuts} = 1 \quad \text{at every depth.} \qquad \text{[std]}$$

The cut counts $1, 3, 7, 15, 31, 127, \ldots$ are the Mersenne numbers; $3, 7, 31, 127$ are prime. Division by 2 alone produces new primes as the count of its own seams. The leftover 1 is the whole, which belongs to no cut. [PG: forced consequence of honest counting]

5. Twins, kept unreduced

For a value and its inverse twin, keep the construction in order: $\frac{4}{3} // \frac{3}{4} \mapsto \frac{3\cdot4}{4\cdot3} = \frac{12}{12}$ and $\frac{3}{2} // \frac{2}{3} \mapsto \frac{2\cdot3}{3\cdot2} = \frac{6}{6}$. Both are the unit; they are not the same object, since each carries its size. Then

$$\frac{12/12}{6/6} \;\mapsto\; \frac{12}{6} = 2 \quad \text{exactly.} \qquad \text{[PG]}$$

Reducing first gives $(4/3)^2 = 16/9$ and $(3/2)^2 = 9/4$, which only bracket 2 to within a factor $9/8$. The bracket is the price of reduction, not a property of 2 and 3. The ten distinct non-unit ratios of $\{1, 2, 3, 4\}$ form five twin pairs about $\varnothing$, the fixed point of $x \mapsto 1/x$. [std] That ten equals $1+2+3+4$ is a coincidence, not structure: the count is $2\sum_{q\le n}\varphi(q)$, which matches the triangular number only at $n = 3, 4$ (at $n = 5$: 18 against 15). [run]

6. The inverse twin turns the hard term into a counter

Proposition 1 (std). *If $\dot f = f^2/2$ then $g = 1/f$ satisfies $\dot g = -\frac{1}{2}$: a rigid translation. If $\dot A = -A^2$ for a matrix A, then $\frac{d}{dt}A^{-1} = I$.*

So $f(t) = 2f_0/(2 - tf_0)$, a Möbius map on Gaussian rationals, and the singular time is $T^* = 2/f_0$ where f_0 is real: a quotient of integers. The peak, the shrinking core and the spin-up are the shadow, through the reciprocal, of a shift. Nothing stretches; two registers trade against a conserved product (skater: $I\Omega = L$; halve one, double the other).

$\max\lvert g(t) - g(0) + t/2\rvert$ at $t = 1$	$\Delta t = 1/64$	$1/128$	$1/256$
$\nu = 0$	6.7×10^{-3}	3.4×10^{-3}	1.7×10^{-3}
$\nu = 1/64$	2.87×10^{-2}	2.89×10^{-2}	—

[run: reciprocal.py; coefficients stepped, f evaluated at exact rational points of the circle]

Without damping the deviation halves with the step: pure scheme error, limit 0. With damping it does not: damping is the only thing that deforms the translation.

7. Damping as one rational per harmonic

The exact decay $e^{-\nu k^2 t}$ is transcendental and stays out of the loop. Two declared rules:

$$\text{implicit: } b_k \mapsto \frac{b_k}{1 + \nu\Delta t\, k^2}, \qquad \text{explicit dyadic: } b_k \mapsto b_k\,(1 - \nu\Delta t\, k^2).$$

Heavy teeth damp harder: larger k, stronger brake. In exact arithmetic the implicit rule is a bijection: multiply by $1 + \nu\Delta t k^2$ to rewind. The damped information is not destroyed; it is stored, and storage grows linearly with steps. [std; run]

Unbounded exact rationals, implicit rule	∼3,000 bits/step, quadratic total time
Unbounded exact rationals, dyadic rule	∼650 bits/step
Typed kernel v2 (§0), all cases	50–330 ms per run; matches exact values to all 6 printed digits
Explicit rule at $\nu\Delta t K^2 = 1$	top harmonic annihilated: exact zero, logged by $\varnothing$ sentinel (stability edge)

Case	rung crosses 1 at	ρ band at end
$\nu = 0$ (exact answer $T^* = 2$)	$t = 132/64 = 33/16$ (first-order step)	$[1.221032, 1.229715]$ at $t = 2.5$
$\nu = 1/1024$	$t = 134/64$	—
$\nu = 1/16$	not by $t = 10$	$[0.793, 0.856]$, still rising

[run: rung_kernel_v2.mjs; $K = 32$ harmonics, band $K/4..3K/4$]

8. Three dimensions: the removed centre

Proposition 2 (std)**.** *Every rotation about c fixes c. Restricted to the centre, the whole rotation group acts as the identity.*

The centre is therefore not a value but $\varnothing$: it has no inside or outside and carries no rotation to measure. [PG] Removing it turns the punctured plane into a cylinder, angle $\times$ scale; with integer rungs, scale is an axis stepped by a counter, not by a logarithm. Removing the axis (not a point) is what gives loops winding in 3D; in 4D one removed point suffices (unit quaternions, Hopf fibres). [std]

Torus anatomy. $1D'$: the empty axis, the reference. $1D'''$: the large rotation about it. $1D''$: the small rotation about the tube's core circle. The two missing centres, axis and core circle, form a Hopf link. The flat torus $\mathbb{R}^3/\mathbb{Z}^3$ decomposes every field into harmonics on integer wavevectors: the integers are built into the shape. [std]

9. Three dimensions: the octant planes

Three planes through $\varnothing$ cut space into eight octants. Each plane carries its own $\{1, i, -1, -i\}$. Composing quarter-turns gives exactly the 24 rotations of the cube: signed permutation matrices of determinant $+1$, pure lookup, closing exactly. [std] $45°$ steps never close and import $\sqrt{2}$; for an exact fine angle use $(5 + 12i)/13$, whose angle is an irrational fraction of a turn (Niven), so it never repeats and undoes exactly.

Per-plane damping. Declare the damping rule as the product of per-plane factors,

$$b_{\mathbf{k}} \mapsto \frac{b_{\mathbf{k}}}{(1 + \nu\Delta t\, k_x^2)(1 + \nu\Delta t\, k_y^2)(1 + \nu\Delta t\, k_z^2)},$$

which equals the isotropic rule up to $O(\Delta t^2)$ and is itself exact, rational, invertible and plane-separable: each plane runs the 1D kernel of §3–7. [std] What does *not* separate is the coupling between planes, which is where the two registers of §6 trade.

10. Fluids as conserved identities

One velocity; every identity (coffee, milk) is its own label field riding on it. Stirring by a volume-preserving map is a bijection on labels, hence reversible; only diffusion merges labels. [std] In pure mathematics identity is conserved: $A = A$, $B = B$, $A \neq B$, and a mixture is a map of where each went, never a new substance. [PG] Physically: Taylor's glycerin demonstration, where a dye blob stirred into a ring reassembles when the stirring is reversed. The two handednesses (clockwise and counter-clockwise, or left- and right-helicity) are separate labels in the same sense.

Chaos in a closed exact system. Any deterministic map on a finite state space is eventually periodic. The doubling map $x \mapsto 2x \bmod 1$, chaotic on $\mathbb{R}$, is periodic on every rational; on

denominator 13 it walks the twelve nonzero residues and closes. Chaos needs infinite resolution: the transcendental import. [std]

11. Open knobs and pre-registrations

1. Sweep K and the band as consts (as γ was swept on the carillon). Confirm: the $\nu = 1/16$ crossing appears as K grows. Kill: ρ saturates below 1 independently of K. [open]
2. Run `hinge_test.py` unchanged on a real power-on recording (scope capture or SDR key-up) to measure real rewind depth. [open]
3. Magnon onset: time-resolved imaging through threshold shows amplitude at the rim first, moving inward (confirm) or nucleating centrally (kill). [open]
4. T1: store each channel's offset from the shared cycle at its hinge, rather than overwrite it, before T1 is run. [open]
5. 3D kernel: three planes, product damping, labels per identity; count distinct interned profiles per rung. Confirm: saturation. Kill: unbounded growth at fixed box. [open]

Scripts: `hinge_test.py`, `viscous_rung.py`, `rung_kernel.mjs`, `rung_kernel_v2.mjs`, `reciprocal.py`. All numbers above were produced on 24 Sep 2026 in a sandboxed container and can be regenerated from those scripts.